

Wei-Yun Lai

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Positions & Education

Year	Position / Degree
2023–present	Postdoctoral research fellow, Institute for Genomic Diversity, Cornell University <i>Advisor: Edward Buckler</i>
2018–2023	Ph.D., Vienna Graduate School of Population Genetics, Univ. of Veterinary Medicine Vienna <i>Advisor: Christian Schlötterer</i>
2017–2018	Research Assistant, Institute of Information Science, Academia Sinica <i>Advisor: Huai-Kuang Tsai</i>
2015–2017	M.Sc., Department of Agronomy, National Taiwan University <i>Advisor: Li-Yu Liu</i>
2011–2015	B.Sc., Department of Agronomy, National Chung Hsing University

Publications

Peer-reviewed Publications

1. Zhai, J., Gokaslan, A., Schiff, Y., Berthel, A., Liu, Z-Y., **Lai, W-Y.**, Miller, Z.R., Scheben, A., Stitzer, M.C., Romay, M.C., Buckler, E.S., & Kuleshov, V. (2025). **Cross-species modeling of plant genomes at single-nucleotide resolution using a pretrained DNA language model.** *Proceedings of the National Academy of Sciences*, 122(24). <https://doi.org/10.1073/pnas.2421738122>
2. **Lai, W-Y.**, Hsu, S-K., Futschik, A., & Schlötterer, C. (2025). **Pleiotropy increases parallel selection signatures during adaptation from standing genetic variation.** *eLife*, 13. <https://doi.org/10.7554/elife.102321.3>
3. Hsu, S-K., **Lai, W-Y.**, Novak, J., Lehner, F., Jakšić, A.M., Versace, E., & Schlötterer, C. (2024). **Reproductive isolation arises during laboratory adaptation to a novel hot environment.** *Genome Biology*, 25(1). <https://doi.org/10.1186/s13059-024-03285-9>
4. **Lai, W-Y.**, Nolte, V., Jakšić, A.M., & Schlötterer, C. (2024). **Evolution of Phenotypic Variance Provides Insights into the Genetic Basis of Adaptation.** *Genome Biology and Evolution*, 16(4). <https://doi.org/10.1093/gbe/evae077>
5. **Lai, W-Y.**, Otte, K.A., & Schlötterer, C. (2023). **Evolution of Metabolome and Transcriptome Supports a Hierarchical Organization of Adaptive Traits.** *Genome Biology and Evolution*, 15(6). <https://doi.org/10.1093/gbe/evad098>
6. Christodoulaki, E., Nolte, V., **Lai, W-Y.**, & Schlötterer, C. (2022). **Natural variation in Drosophila shows weak pleiotropic effects.** *Genome Biology*, 23(1). <https://doi.org/10.1186/s13059-022-02680-4>
7. **Lai, W-Y.**, & Schlötterer, C. (2021). **Evolution of phenotypic variance in response to a novel hot environment.** *Molecular Ecology*, 31(3), 934-945. <https://doi.org/10.1111/mec.16274>
8. Huang, J-H., Liao, Y-R., Lin, T-C., Tsai, C-H., **Lai, W-Y.**, Chou, Y-K., Leu, J-Y., Tsai, H-K., & Kao, C-F. (2021). **iTARGETX analysis of yeast deletome reveals novel regulators of transcriptional buffering in S phase and protein turnover.** *Nucleic Acids Research*, 49(13), 7318-7329. <https://doi.org/10.1093/nar/gkab555>

Preprints

1. Hsu, S-K., Schulz, A.J., Hale, C.O., Costa-Neto, G., Miller, Z.R., Stitzer, M.C., Wrightsman, T., Zhai, J., **Lai, W-Y.**, Dawson, H.D., & *et al.* (2026). **Convergent genome- and gene-level constraints shape repeated environmental adaptation in grasses.** *bioRxiv*. <https://doi.org/10.64898/2026.06.01.729361>

2. Epstein, R., Wheeler, J., Hubisz, M., Sun, Q., Bukowski, R., Zhai, J., Lai, W-Y., Buckler, E., & Pawlowski, W.P. (2024). **The maize recombination landscape evolved during domestication.** *bioRxiv*. <https://doi.org/10.1101/2024.11.04.621928>

Presentations

- 2026 **Pan-Genomics and GWAS Unveils the importance of ancient whole genome duplication for maize environmental adaptation.** *Poster presentation with lightning talk*, The 68th Annual Maize Genetic Meeting, February 26 - March 1, Cologne, Germany
- 2026 **Pan-Genomics and GWAS Unveils the importance of ancient whole genome duplication for maize environmental adaptation.** *Oral presentation*, Corn Breeding Research Meeting, February 25 - 26, Cologne, Germany
- 2025 **Integreting AI and Pan-genome to identify adaptive variants in maize.** *Oral presentation*, ACSAC11, May 19-21, Taipei, Taiwan
- 2025 **Integreting AI and Pan-genome to identify adaptive variants in maize.** *Poster presentation*, The 67th Annual Maize Genetic Meeting, March 6-9, Union Station in St. Louis, Mo, USA
- 2024 **Building a practical haplotype graph from PacBio assemblies for genotype imputation and allele mining in maize.** *Oral presentation*, PacBio seminar, November 12, Cornell University, NY, USA
- 2024 **Identify causal variants in a maize population through a DNA language model: PlantCAD.** *Poster presentation*, ICQG7, July 22-26, Vienna, Austria
- 2024 **Mining cold-tolerance alleles in maize.** *Poster presentation*, The 66th Annual Maize Genetic Meeting, February 29-March 3, Raleigh, NC, USA
- 2023 **The evolution of aging transcriptome in Drosophila.** *Poster presentation with lightning talk*, GLAM-Evogen 2023, August 19, Cornell University, NY, USA
- 2022 **Pleiotropy is associated with the parallel gene expression evolution during adaptation to a novel environment.** *Oral presentation*, PEQG22, June 7-10, Pacific Grove, CA, USA
- 2021 **Evolution of phenotypic variance provides insights into the genetic basis of adaptation.** *Oral presentation*, EvolVienna, January 4-6, Online
- 2021 **Evolution of phenotypic variance provides insights into the genetic basis of adaptation.** *Oral presentation*, Popgroup54, January 4-6, Online
- 2020 **High stability of gene expression variance in Drosophila simulans populations evolving in a novel high-temperature environment.** *Poster presentation*, TAGC, April 22-25, Online
- 2017 **Study of alternative splicing events in maize transcriptome under abiotic stress.** *Poster presentation*, Plant and Animal Genomes 25, January 14-18, San Diego, CA, USA

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